

FRONTIERS AND PROGRESS IN QUANTUM MEASUREMENT: LEARNING-ASSISTED NITROGEN-VACANCY MAGNETOMETRY

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ABSTRACT

Nitrogen-vacancy centers in diamond provide an optical interface for measuring magnetic fields through spin resonance. This course report reviews the physical sensing mechanism, measurement protocols, machine-learning inversion, and adaptive experiment design, emphasizing the distinction between extracting spectral parameters and measuring a calibrated field. A reproducible simulation study illustrates these issues through field-strain identifiability, neural and likelihood estimators, a nine-level electron-nuclear model, uncertain bias gain, and repeated photon-budget comparisons. The results show that observation design and model fidelity can matter more than a reconstruction loss, and that adding identifiable nuisance parameters does not always improve finite-budget accuracy. An audit of public ODMR sequences identifies temporally structured noise without supplying magnetic-field ground truth. The report combines literature synthesis, analytical examples, and executed experiments to explain both the opportunities and practical limits of learning-assisted quantum magnetometry.

Index Terms— Quantum sensing, nitrogen-vacancy centers, identifiability, model mismatch, adaptive measurement.

1. INTRODUCTION AND REPORT SCOPE

Quantum measurement connects a controllable quantum system to an observable physical quantity [1]. Nitrogen-vacancy (NV) centers in diamond offer an optically accessible spin system for magnetic sensing at room temperature, with applications in local field detection, nanomagnetism, and the life sciences [2, 3]. Understanding this platform requires linking microscopic spin physics to the acquisition and calibration processes that produce a useful measurement. It also makes the measurement chain concrete: a quantum spin carries field information, while optical instrumentation and statistical inference determine how that information is acquired, calibrated, and interpreted.

The field affects spin-transition frequencies, microwave excitation probes these transitions, and spin-dependent fluorescence supplies the readout (Fig. 1). The spectrum is an intermediate observable whose interpretation depends on a response model and calibration. Photon collection, contrast, and spin dephasing also constrain precision [4]. Progress consequently involves coordinated improvements in materials, control, readout, and inference, rather than a single accuracy metric.

Three questions organize this report: does the observation distinguish the field from nuisance parameters, how does model mismatch affect recovery, and how should a finite measurement budget be distributed? The equal-dip axial example illustrates field-strain ambiguity; hyperfine structure and uncertain bias gain test model fidelity. Together, these issues connect sensor principles to identifiability, calibration, and experimental design.

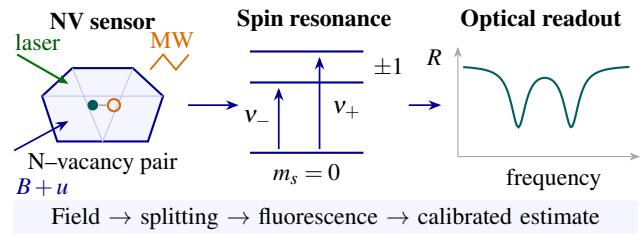


Fig. 1. NV measurement chain. Microwave (MW) transitions are read optically; calibration converts spectral information to a field. The drawing is conceptual.

Learning can supply direct parameter predictions or initial estimates for physical refinement; embedded NV magnetometry illustrates its practical use [5]. Adaptive acquisition instead selects measurement settings from accumulated observations [6, 7]. Neither approach makes calibration unnecessary. Their comparison should therefore state what is measured, which assumptions are shared, and whether photon budgets are matched.

For the Sensor Principles and Applications course, this report selects frontiers and progress in quantum measurement. It combines a focused narrative review, analytical examples, and executed simulations of axial NV magnetometry. Figure 2 connects acquisition, neural estimation, physical refinement, and evaluation. The experiments examine distribution shift, a nine-level simulator, bias calibration, and repeated adaptive measurements. A public-data audit separately checks noise without assuming magnetic-field ground truth. This scope supports a substantial computational study without NV equipment while keeping conclusions tied to available evidence.

2. REVIEW OF NV MEASUREMENT AND LEARNING METHODS

2.1. NV sensing principles and measurement protocols

An NV defect consists of a substitutional nitrogen adjacent to a vacancy in diamond. The negatively charged center has a spin-triplet ground state; its zero-field splitting is approximately 2.87 GHz near room temperature. Optical excitation preferentially prepares the $m_s = 0$ state, while spin-dependent nonradiative pathways make the fluorescence depend on spin population. Sweeping a microwave frequency through a spin transition therefore produces a fluorescence dip. ODMR connects this optical signal to the spin Hamiltonian, rather than measuring the magnetic field directly [2, 3].

The conversion from a spectrum to a field requires more than locating a dip. Strain and electric fields affect low-field splittings, temperature changes the zero-field splitting, and microwave drive and optical power affect the observed linewidth and contrast. Bias fields and independent calibration help distinguish these effects. Conse-

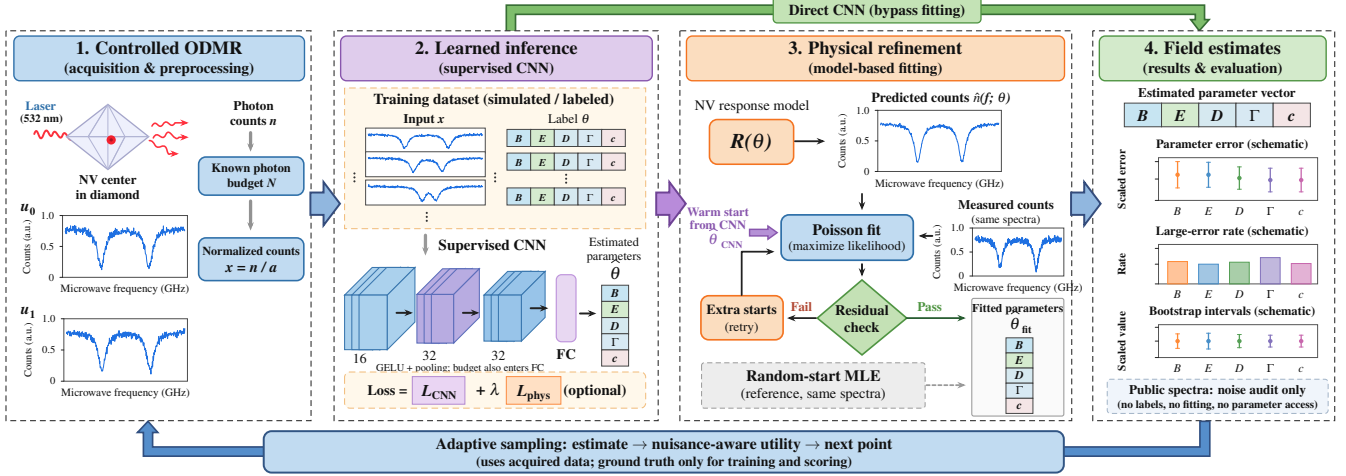


Fig. 2. Learning-assisted NV magnetometry. The CNN provides direct predictions or a likelihood warm start; a residual gate can trigger extra starts. Adaptive sampling is a separate measurement protocol. All spectra and evaluation glyphs in this framework are schematic; quantitative results appear in Section 5.

quently, accurate spectral reconstruction need not imply an accurate absolute-field estimate. This distinction motivates the identifiability analysis in Section 3 and prevents interpreting every frequency shift as a magnetic signal [4, 8].

Continuous-wave ODMR applies optical illumination and microwave excitation during readout, making frequency sweeps conceptually simple and suitable for the simulation below. Pulsed protocols separate preparation, coherent evolution, and readout: Ramsey interrogation encodes a static or slowly varying field in accumulated phase, whereas echo and related decoupling sequences emphasize selected time-dependent signals. These protocols use different information and have different bandwidth and coherence constraints; conclusions from a static ODMR sweep should not be transferred to all quantum sensing experiments [1, 3].

Single-center measurements provide a localized probe, while ensembles can increase the optical signal and support imaging. Ensemble orientation, inhomogeneous broadening, and collection efficiency complicate interpretation. Sensitivity depends on spin coherence, contrast, photon collection, and acquisition time, not only on the fitted frequency error [4]. Widefield microscopy further couples spatial resolution to optical acquisition and reconstruction choices [8]. The present benchmark isolates axial spectral inversion and fixed photon allocation, leaving these device-specific engineering tradeoffs outside its quantitative claims.

2.2. Learning-assisted inference, adaptation, and reliability

Conventional analysis fits a physical response model to each spectrum. Its advantages are explicit parameters and a clear connection to calibration, but overlapping resonances and poor initialization can create difficult optimization problems. Supervised learning instead amortizes inference: training is costly, while a trained network maps new spectra to parameters in one forward pass. Homrighausen et al. demonstrate a standalone embedded implementation using experimentally acquired ODMR spectra from randomly oriented NV ensembles [5]. This illustrates practical deployment; its calibration and prediction target remain specific to the demonstrated sensor.

Yao et al. study direct CNN prediction, uncertainty estimation, and a hybrid strategy in which network outputs initialize

spectral fitting [9]. Synthetic and experimental validation are both considered, including thermometry and magnetic imaging. Their high-signal reference spectra must be distinguished from independently known magnetic-field truth. Physics-guided calibration [10] and meta-learning for control [11] extend the discussion toward transfer and protocol adaptation, but calibration, spectral inversion, and microwave control are different tasks. Domain randomization broadens simulated training variation [12]; it cannot by itself establish robustness to an unrepresented noise process or eliminate a physical ambiguity.

Adaptive design changes what is measured in response to acquired data. Sequential Bayesian ODMR methods update a parameter distribution and select informative instrument settings [6]. Kelley and McMichael further account for unequal parameter importance and measurement expense [7]. Their framework clarifies why reducing uncertainty in every parameter equally can waste effort when only the field is of interest, and why information per measurement differs from information per unit experimental cost. The local Fisher rule used here illustrates this principle without reproducing the full Bayesian algorithms.

Multiple bias settings can also supply information absent from a single spectrum. Existing work addresses joint magnetic-field and sensor-orientation recovery [13]; our example assumes a known axis and asks a narrower calibration question. Reliability requires reporting the training domain, physical assumptions, reference labels, failures, and uncertainty calibration. Distribution-free prediction intervals require appropriate exchangeability conditions [14]; they do not automatically cover a new sensor or a shifted observation model. Taken together, the literature suggests evaluating acquisition, inference, and calibration as one measurement chain.

3. OBSERVATION MODEL AND IDENTIFIABILITY

Let B denote the signed field component along a known NV axis, u a controlled bias, D the zero-field splitting, and $E \geq 0$ an effective transverse strain/electric-field parameter. Neglecting hyperfine coupling, the spin-1 Hamiltonian is

$$H/h = DS_z^2 + E(S_x^2 - S_y^2) + \gamma_e(B + u)S_z, \quad (1)$$

where $\gamma_e = 28 \text{ MHz/mT}$. The two resonance frequencies are

$$v_{\pm}(u) = D \pm \sqrt{\gamma_e^2(B+u)^2 + E^2}. \quad (2)$$

We initially assume equal Lorentzian widths and contrasts, with parameter vector $\theta = (B, E, D, \Gamma, c)$:

$$R(f, u; \theta) = 1 - \sum_{s \in \{-, +\}} \frac{c}{1 + [(f - v_s(u))/\Gamma]^2}. \quad (3)$$

Here Γ is the half width at half maximum and c is the contrast of each dip. Counts satisfy $n_{ij} \sim \text{Poisson}(a_{ij}R_{ij})$. We define the nominal photon budget as $N = \sum_{ij} a_{ij}$, the expected off-resonance count summed over all dwell allocations. Realized counts are random. This is a photon-allocation comparison, not a calibrated hardware energy or sensitivity measurement.

Proposition 1 (calibrated biases). Under (3), a single bias does not generally identify both B and E . Two distinct known biases identify the signed axial field from noiseless resonance positions when B, E, D remain unchanged between probes.

Derivation. Define $r_i = [(v_+(u_i) - v_-(u_i))/2]^2$. A single measurement determines D and $r_i = \gamma_e^2(B + u_i)^2 + E^2$, leaving a continuum of (B, E) pairs. Subtracting two such equations gives

$$B = \frac{r_1 - r_0 - \gamma_e^2(u_1^2 - u_0^2)}{2\gamma_e^2(u_1 - u_0)}. \quad (4)$$

Then $E^2 = r_0 - \gamma_e^2(B + u_0)^2$ and $D = (v_+ + v_-)/2$. This noiseless result does not guarantee stability for overlapping peaks or small bias separation. Additional amplitude or hyperfine information may change the equal-dip ambiguity.

Suppose the applied field is au with unknown positive gain a . The squared splitting becomes

$$r(u) = \alpha u^2 + \beta u + \eta, \quad \alpha = \gamma_e^2 a^2, \quad \beta = 2\gamma_e^2 a B. \quad (5)$$

The constant is $\eta = \gamma_e^2 B^2 + E^2$. Two biases cannot generally recover the three coefficients. Three distinct biases recover the noiseless quadratic, yielding $a = \sqrt{\alpha}/\gamma_e$ and $B = \beta/(2\gamma_e^2 a)$. If an unknown additive bias offset b_0 is also present, only $B + b_0$ is observable: no number of such probes separates them without an external reference. This separates identifiable gain correction from an unidentifiable absolute offset. More parameters may also increase finite-budget variance, as evaluated below.

4. ESTIMATORS AND MEASUREMENT DESIGN

4.1. Learning-assisted estimation and model fidelity

The input comprises two normalized count spectra, $x_{ij} = n_{ij}/a_{ij}$, and $\log_{10} N - 6$. A 1D-CNN uses convolutional channels 16, 32, and 32, followed by a width-128 hidden layer and five outputs. The two-bias model has approximately 0.173 million parameters. Targets are normalized using their training centers and half ranges. Let z denote this normalized parameter vector. The physics-guided variant minimizes

$$\mathcal{L} = \text{MSE}(\hat{z}, z) + 0.03 \bar{d}, \quad (6)$$

$$d_{ij} = \frac{2}{0.08^2} \left[\hat{R}_{ij} - x_{ij} + x_{ij} \log \frac{x_{ij}}{\hat{R}_{ij}} \right]. \quad (7)$$

The bar averages observations and samples. This normalized-response divergence is not count-weighted; responses are clamped positive. It does not guarantee unique or valid parameters.

We compare direct CNN prediction, the regularized CNN, network-initialized likelihood fitting, and a residual-gated refinement. Classical Poisson maximum likelihood uses one or four random initializations. All refinements use the same broader bounds: $B \in [-0.3, 0.3] \text{ mT}$, $E \in [0.02, 8] \text{ MHz}$, $D \in [2863, 2877] \text{ MHz}$, $\Gamma \in [0.35, 7] \text{ MHz}$, and $c \in [0.005, 0.25]$. The gated method adds three starts above the approximate 99% χ^2_{317} deviance threshold. Boundary effects and misspecification limit this diagnostic.

To test physical model mismatch beyond an artificially shifted triplet, we diagonalize a nine-level $S = 1, I = 1$ ground-state Hamiltonian:

$$H_9/h = H/h + A_{\parallel} S_z I_z + A_{\perp} (S_x I_x + S_y I_y) + P I_z^2 - \gamma_n (B + u) I_z. \quad (8)$$

We use $A_{\parallel} = -2.16$, $A_{\perp} = -2.70$, and $P = -4.95 \text{ MHz}$, with $\gamma_n = 0.003077 \text{ MHz/mT}$ [15]. Transition weights average two transverse microwave polarizations and an unpolarized nuclear population; the summed oscillator strength is normalized at each bias. Common Lorentzian linewidth and contrast remain assumptions. This is a ground-state spectral simulator, not a full optical pumping or saturation model.

Matched CNNs retain the architecture and training size. Four-start matched and misspecified likelihood fits share observations; zero hyperfine terms numerically recover the two-line response.

4.2. Adaptive measurement design

Each adaptive trial spends 25% of the budget on 32 coarse points and the remaining 75% on 64 points in eight updates. Candidates consist of 129 frequencies at each of two biases. For gradients with respect to normalized parameters, we approximate the local information by

$$J = 3I + \sum_k (a_k/R_k) g_k g_k^{\top}. \quad (9)$$

This regularizer is a classical design aid. With $C = J^{-1}$, the local reduction of the B coordinate variance from a candidate is

$$\Delta_B = \frac{(e_B^{\top} C g)^2}{R/a + g^{\top} C g}. \quad (10)$$

We compare uniform sampling, maximizing ag_B^2/R alone, and maximizing (10). All rules use acquired data, shared restart mechanisms, and equal nominal budgets. They illustrate local design rather than full Bayesian inference [6, 7].

5. EXPERIMENTS AND RESULTS

5.1. Benchmark setup, identifiability, and distribution shift

Training ranges are $B \in [-120, 120] \mu\text{T}$, $E \in [0.5, 4] \text{ MHz}$, $D \in [2868, 2872] \text{ MHz}$, $\Gamma \in [1, 3] \text{ MHz}$, and $c \in [0.04, 0.12]$. Baseline biases are 0 and 250 μT , with 161 frequencies per bias over 2854–2886 MHz. Photon budgets are 10^5 , 10^6 , and 10^7 . Each of three training seeds supplies 20,000 training and 2,000 validation samples; validation selects a checkpoint over 22 epochs. There are 12 training runs, including three nine-level CNNs. Shared simulator/seed configurations reuse training data.

Fixed sweeps allocate counts equally. Test-case bootstrap preserves paired predictions and neural seeds; adaptive resampling is stratified by independent batch. Intervals condition on saved models and simulated cases. Original MLE results retain only aggregate

Table 1. Original test: field RMSE (μT) at $N = 10^6$; neural entries average three seeds. HF: shifted triplet.

Method	ID	D shift	E shift	HF
Single-bias CNN	68.39	68.45	68.54	67.72
CNN	7.10	31.96	20.37	11.94
Physics CNN	7.09	31.99	20.29	11.91
NN + fit	5.90	10.71	6.62	11.26
Gated NN + fit	5.90	4.79	6.62	11.25
MLE: 1 start	5.90	10.02	6.62	11.25
MLE: 4 starts	5.90	4.79	6.62	11.26

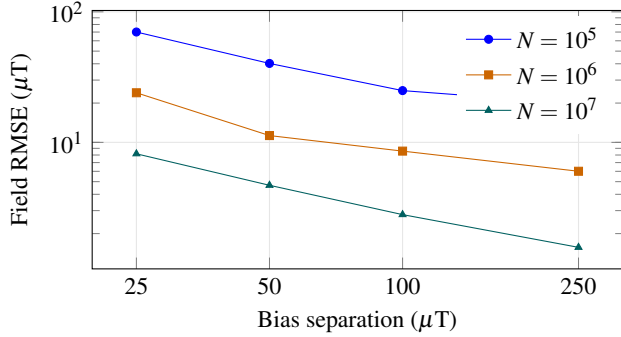


Fig. 3. Bias separation with equal photon budgets: 128 shared parameter vectors, four-start MLE.

errors. Independent ID calibration uses 1,500 cases at $N = 10^6$ to check 95% residual intervals, without OOD guarantees [14]. Each experiment is discussed together with its results.

The original test uses 192 cases per budget/condition: ID, $D + 3.5$ MHz, $E + 2.5$ MHz, $\Gamma + 2$ MHz, a shifted triplet, and baseline slope. Table 1 summarizes four conditions. The single-bias ablation discards the second spectrum and its counts. The single-bias signed-field error is near the symmetric prior standard deviation, consistent with its sign ambiguity. Two-bias CNN prediction is much more accurate, but adding the reconstruction loss changes ID RMSE only marginally. Under zero-field-splitting shift, gated refinement matches the strong multistart baseline. This improvement combines a wider physical search range with optimization restarts, and is not evidence of a new architecture. The shifted-triplet case leaves appreciable error even after refinement.

The matched-budget spacing sweep uses 128 shared parameter vectors at 25, 50, 100, and 250 μT separation to test recovery stability. At $N = 10^6$, four-start fitting yields RMSEs of 23.99, 11.27, 8.57, 6.01 μT for increasing separation. The two smallest spacings incur a substantial variance/optimization penalty. Larger separation helps within the tested frequency range, but this is not an unlimited prescription: a larger bias can move transitions out of the sweep or introduce hardware costs.

5.2. Physical-model fidelity and bias calibration

For each photon budget, 192 independent vectors generate nine-level test spectra (Table 2). Transfer CNNs use the two-line training model; matched CNNs use the nine-level model. Both likelihood fits use four starts on identical observations. At $N = 10^6$, the two-line likelihood RMSE is 11.21 μT , compared with 8.06 μT for the matched fit; the newly trained CNN gives 8.76 μT . These results test model fidelity directly, rather than treating all hyperfine structure as uniform additive line offsets. A stronger simulator does not

Table 2. Nine-level test: field RMSE (μT), 192 cases; four-start MLE.

Method	10^5	10^6	10^7
Transfer CNN	29.16	11.21	7.11
Transfer physics CNN	29.16	11.22	7.13
Matched CNN	27.71	8.76	4.49
Two-line MLE	35.90	11.21	5.02
Nine-level MLE	31.46	8.06	2.36

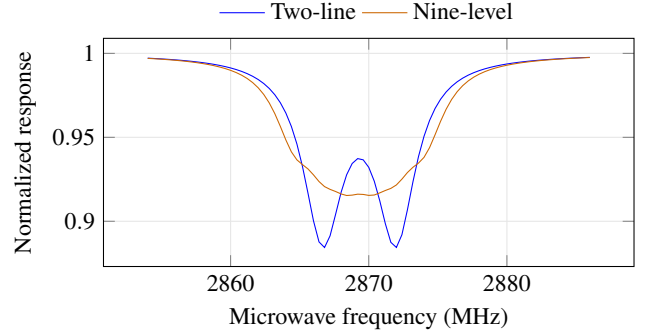


Fig. 4. Two-line and nine-level responses at zero bias for identical parameters.

Table 3. Bias-gain test: field RMSE (μT); shared observations within each protocol. Oracle rows use true gain.

Protocol	10^6	10^7
2 biases, unit gain	8.78	7.32
3 biases, unit gain	7.67	6.08
3 biases, fitted gain	17.65	5.53
2 biases, oracle gain	5.69	1.78
3 biases, oracle gain	5.45	1.70

eliminate assumptions about nuclear populations, drive polarization, linewidth, or optical dynamics.

The gain test uses 192 shared vectors with $a \sim \mathcal{U}[0.95, 1.05]$. Two- and three-bias protocols share total N ; three-bias fits share observations. Oracle rows receive the true gain, an independently calibrated reference rather than a deployable estimate (Table 3). At $N = 10^6$, its RMSE is 17.65 μT versus 7.67 μT when three-bias fitting fixes unit gain. The oracle quantifies what independent gain calibration could recover on the same observations. Identifiability alone therefore does not imply a finite-budget improvement. Practical error combines statistical and optimization effects, supporting external gain calibration or informative constraints at limited counts.

5.3. Adaptive measurement and public-data audit

Three independent batches of 64 vectors per budget/rule yield 1,728 adaptive trials, with all estimates and actions saved. Table 4 reports the repeated comparison. At $N = 10^6$, uniform and nuisance-aware sampling have pooled RMSEs 12.12 and 6.59 μT . The paired bootstrap interval for the difference, uniform minus nuisance-aware, is [4.04, 7.08] μT . Saved results retain the low-budget outcome and error-above-30- μT rates. An inaccurate coarse estimate can misdirect later sampling, so failure cases remain important.

The budget excludes switching and reference-count costs. Full Bayesian design and hardware cost measurements would extend this comparison.

The Figshare record [16] contains 4,693 sweeps and 311 fre-

Table 4. Adaptive trials: pooled RMSE (μT); final column: stratified 95% interval at $N = 10^6$.

Rule	10^5	10^6	10^7	95% interval
Uniform	41.80	12.12	4.41	[10.39, 13.84]
B only	45.74	14.18	5.00	[10.38, 18.86]
Nuisance-aware	34.75	6.59	2.22	[5.55, 7.54]

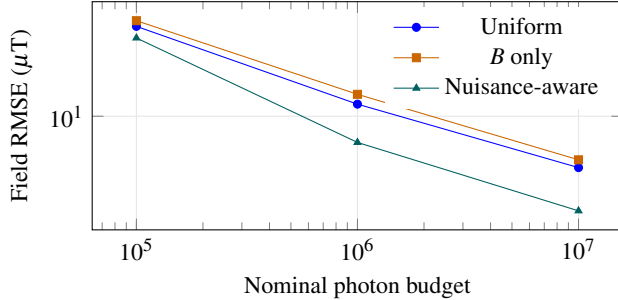


Fig. 5. Adaptive trial RMSE (192 cases/budget). At 10^5 , the paired uniform-minus-aware 95% interval includes zero.

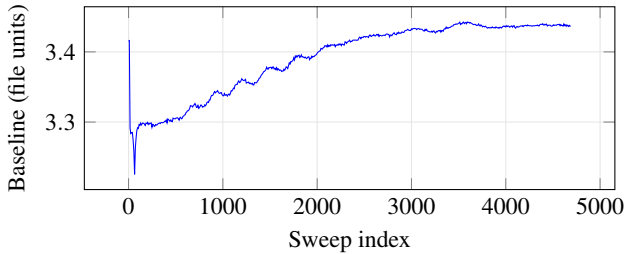


Fig. 6. Off-resonance baseline of public sweeps [16]; every eighth point shown.

quencies. We use it only for spectral and noise auditing: there is no per-sweep magnetic-field ground truth or matching calibrated two-bias protocol. The file contains floating-point records and an analog acquisition-channel identifier, so its unit label alone does not establish independent integer Poisson counts. The mean of the first 20 off-resonance frequency bins has lag-one correlation 0.999. PCA is centered and fitted on the first 3,754 sweeps; the first component explains 98.0% of training variance, and the remaining 939 sweeps are projected without refitting. This shows temporally structured variation, not its unique physical cause.

The audit reveals a mismatch with the independent Poisson simulator, rather than validating field accuracy or sensor sensitivity. Other assumptions include known axial orientation, stable biases, fixed hyperfine constants, three neural seeds, and local design approximations. Together, the public audit and controlled simulations motivate repeated reference-field measurements as the next step toward calibrated instrument evaluation.

6. DISCUSSION AND CONCLUSION

NV sensing combines quantum-state encoding with classical acquisition and inference. The reviewed methods improve different parts of this chain: optical protocols determine information, neural inference amortizes computation, physical refinement supplies model structure, and adaptive design allocates measurements. The simulations illustrate why these benefits are conditional. A learned prior cannot remove a measurement ambiguity; model fidelity matters un-

der hyperfine shifts; and extra calibration parameters or poorly initialized adaptive actions can increase error.

Future laboratory work should calibrate bias and frequency axes, acquire repeated spectra at reference fields, separate training and testing by session, and count drift, switching, and reference measurements in the acquisition cost. The report connects current literature to executed experiments while distinguishing simulated accuracy from experimentally validated sensitivity.

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